

ORI FRIESEN

MATHEMATICIAN

FRIESEN.37@OSU.EDU
COLUMBUS, OHIO
WWW.ORIFRIESEN.COM

EDUCATION

Fall 2025 – Present	Ph.D. Program in Theoretical Mathematics The Ohio State University, Columbus, OH
May 2025	B.A. in Mathematics, Computer Science Minor, Geospatial Science Minor Macalester College, Saint Paul, MN Thesis: <i>On the Hölder Continuity of the Brascamp-Lieb Constant</i> Overall GPA: 3.96

PAPERS & PREPRINTS

1. O. Friesen, C. Kolko, N. Layman, K. Lorenzen, S. Zaske, and A. Ziegler. "A cospectral construction for the generalized distance matrix" *Special Matrices*, vol. 13, no. 1, 2025, pp. 20250037. <https://doi.org/10.1515/spma-2025-0037>
2. O. Friesen. "On the Hölder Continuity of the Brascamp-Lieb Constant" (2025). *Mathematics, Statistics, and Computer Science Honors Projects*. 98. https://digitalcommons.macalester.edu/mathcs_honors/98
3. O. Friesen and L. Ziegelmeier. "Understanding U.S. Racial Segregation Through Persistent Homology," arXiv preprint [arXiv:2410.10886](https://arxiv.org/abs/2410.10886), submitted to *SIAM Review, Research Spotlights*.

RESEARCH

August 2024 – May 2025	Honors Independent Research at Macalester Research in Functional and Harmonic Analysis
March 2023 – May 2025	Independent Research at Macalester Research in Topological Data Analysis
June 2024 – August 2024	Research Assistant at Iowa State University Math REU Research in Spectral Graph Theory
January 2023 – August 2023	Traffic Analyst Fellow at ReConnect Rondo Through the Chuck Green Fellowship at Macalester College

TEACHING

September 2023 – May 2025	Teaching Assistant and Grader at Macalester Providing individualized support during drop-in office hours, grading homework and other assignments, and facilitating learning by assisting during class sessions.
---------------------------	---

Courses Assisted

- Spring 2025: MATH 471 – Topology
- Fall 2024: MATH 375 – Graph Theory
- Spring 2024: MATH 376 – Algebraic Structures
- Fall 2023: MATH 236 – Linear Algebra

TALKS

January 2025	Joint Mathematics Meetings <i>AMS – PME Undergraduate Student Poster Session, I</i> “Cospectral Constructions for the Exponential Distance Matrix” <i>AMS Special Session on Topological Data Analysis: Theory and Applications, I</i> “Understanding U.S. Racial Segregation Through Persistent Homology”
October 2024	Minnesota GIS/LIS Consortium <i>Undergraduate Student Competition 1</i> “Understanding the Shape of U.S. Racial Segregation Through Persistent Homology”
April 2024	Midwestern Undergraduate Geography Symposium <i>Paper Presentation Session 3C – Urban Geography</i> “On the Path to Knowledge: Exploring aspects of transportation and accessibility to the Hayden Heights Public Library”
January 2024	Joint Mathematics Meetings <i>AMS – PME Undergraduate Student Poster Session, I</i> “Using Persistent Homology to Study the Shape of Racial Segregation in United States Cities”

LEADERSHIP

September 2023 – May 2025	Student Advisory Board Co-Leader Department of Mathematics, Statistics, & Computer Science, Macalester College Bridging the gap between students and professors and facilitating inclusion in the MSCS department through a multitude of initiatives and activities.
---------------------------	---

HONORS AND CERTIFICATIONS

April 2025	Camp Endowed Prize in Mathematics Macalester College
June 2024	Outstanding Student Representative Minnesota GIS/LIS Consortium
Inducted May 2024	Gamma Theta Upsilon (International Geographical Honors Society) Epsilon Kappa Chapter
January 2023 – August 2023	2023 Chuck Green Fellow Department of Political Science, Macalester College